



Gosford High School

2015

TRIAL
HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics Extension 2

- **General Instructions**

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations

Total Marks – 100

Section I Pages 2 – 5

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section
- Answer on the response sheet provided

Section II Pages 6 – 12

90 marks

- Attempt Questions 11 – 16
- Start a **new booklet** for **each** question
- Answer Question 14(a) on the answer sheet provided
- Allow about 2 hours and 45 minutes for section II

Section I

10 Marks

Attempt Questions 1-10.

Allow about 15 minutes for this section.

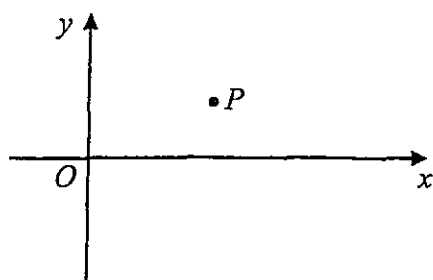
Use the multiple-choice answer sheet for questions 1-10.

-
- 1 What is the value of $\lim_{h \rightarrow 0} \frac{\sqrt{h+1}-1}{h}$? 1
- (A) 0
- (B) $\frac{1}{2}$
- (C) 1
- (D) 2
- 2 A, B, C are three consecutive terms in an arithmetic progression. 1
- Which of the following is a simplification of $\frac{\sin(A+C)}{\sin B}$?
- (A) $2 \cos B$
- (B) $\sin 2B$
- (C) $\cot B$
- (D) 1
- 3 What is the number of asymptotes on the graph of the curve $y = \frac{x^2}{x^2-1}$? 1
- (A) 1
- (B) 2
- (C) 3
- (D) 4

Marks

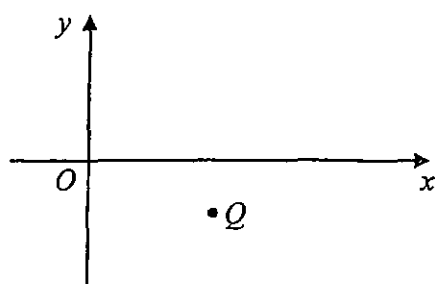
1

4 On the Argand diagram below, P represents the complex number z .

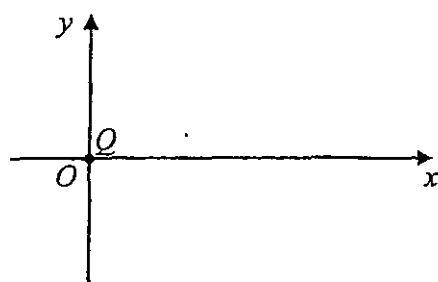


Which of the following Argand diagrams shows the point Q representing $z + \bar{z}$?

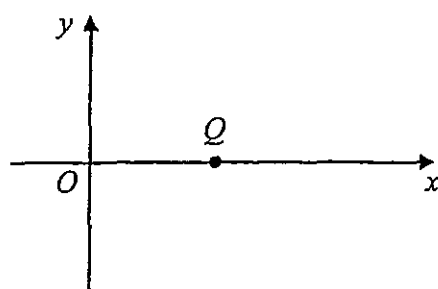
(A)



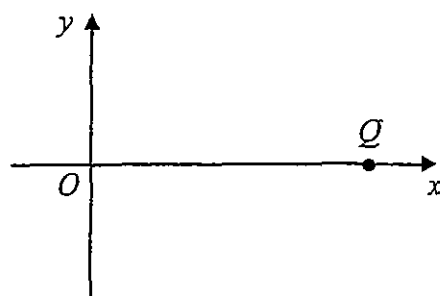
(B)



(C)



(D)



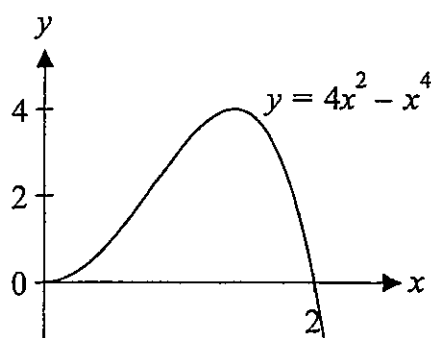
5 What is the acute angle between the asymptotes of the hyperbola $\frac{x^2}{3} - y^2 = 1$? 1

- (A) $\frac{\pi}{6}$
- (B) $\frac{\pi}{4}$
- (C) $\frac{\pi}{3}$
- (D) $\frac{\pi}{2}$

6 Which of the following is an expression for $\int_0^{\frac{\pi}{2}} \frac{1}{1+\sin x} dx$ after the substitution $t = \tan \frac{x}{2}$? 1

- (A) $\int_0^1 \frac{1}{1+2t} dt$
- (B) $\int_0^1 \frac{2}{1+2t} dt$
- (C) $\int_0^1 \frac{1}{(1+t)^2} dt$
- (D) $\int_0^1 \frac{2}{(1+t)^2} dt$

7



The region in the first quadrant bounded by the curve $y = 4x^2 - x^4$ and the x axis between $x=0$ and $x=2$ is rotated through 2π radians about the y axis. Which of the following is an expression for the volume V of the solid formed ? 1

- (A) $V = 2\pi \int_0^4 \sqrt{4-y} dy$
- (B) $V = 4\pi \int_0^4 \sqrt{4-y} dy$
- (C) $V = 8\pi \int_0^4 \sqrt{4-y} dy$
- (D) $V = 16\pi \int_0^4 \sqrt{4-y} dy$

Marks

1

8 The equation $x^4 + px + q = 0$, where $p \neq 0$ and $q \neq 0$, has roots α, β, γ and δ .

What is the value of $\alpha^4 + \beta^4 + \gamma^4 + \delta^4$?

(A) $-4q$

(B) $p^2 - 2q$

(C) $p^4 - 2q$

(D) p^4

9 Which of the following is the range of the function $f(x) = \sin^{-1} x + \tan^{-1} x$?

1

(A) $-\pi < y < \pi$

(B) $-\pi \leq y \leq \pi$

(C) $-\frac{3\pi}{4} \leq y \leq \frac{3\pi}{4}$

(D) $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

10 If $e^x + e^y = 1$, which of the following is an expression for $\frac{dy}{dx}$?

1

(A) $-e^{x-y}$

(B) e^{x-y}

(C) e^{y-x}

(D) $-e^{y-x}$

Section II

90 Marks

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section.

Answer the questions on your own paper, or in writing booklets if provided.

Start each question on a new page.

All necessary working should be shown in every question.

Question 11 (15 marks)

Use a SEPARATE writing booklet

- (a) If $z = 1 + 3i$ and $w = 2 - i$ find in the form $a + ib$ (for real a and b) the values of
- (i) $\bar{z} - w$ 1
 - (ii) zw 1
- (b)(i) Express $-1 + \sqrt{3}i$ in modulus/argument form. 2
- (ii) Hence find the value of $z^8 - 16z^4$ in the form $a + ib$ where a and b are real. 2
- (c) In the Argand diagram $OABC$ is a square, where O, A, B, C are in anti-clockwise cyclic order. The complex number z is represented by the vector $\overrightarrow{OA}$.
- (i) Find in terms of z the complex numbers represented by the vectors $\overrightarrow{OC}$ and $\overrightarrow{OB}$. 2
 - (ii) If the vector $\overrightarrow{OB}$ represents the complex number $4 + 2i$, find z in the form $a + ib$ where a and b are real. 2
- (d) The polynomial $P(x) = x^6 + ax^3 + bx^2$ has a factor $(x + 1)^2$. 3
- Find the values of the real numbers a and b .
- (e) The equation $x^4 + bx^3 + cx^2 + dx + 1 = 0$ has roots $\alpha, \frac{1}{\alpha}, \beta, \frac{1}{\beta}$. 2
- Show that $b = d$

a) Find the following

(i)

$$\int \frac{e^{-x}}{1 + e^x} dx$$

3

(ii)

$$\int \frac{x^2}{x + 1} dx$$

2

b) Find the exact value of the following definite integral:

$$\int_0^{\frac{\pi}{6}} \sec^3 2\theta d\theta$$

4

c) By using the substitution of $x = \tan \theta$, show that

$$\int_1^{\sqrt{3}} \frac{1}{x^2 \sqrt{1 + x^2}} dx = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \operatorname{cosec} \theta \cot \theta d\theta$$

3

d) The area bounded by the line $y = (2 - x)$ and the x axis, is rotated about the y axis. By using the method of cylindrical shells, find the volume generated.

3

End of Question 12

Question 13 (15 marks)**Start a new booklet****Marks**

(a) (i) If $z = \cos\theta + i\sin\theta$, show that $z^n + \frac{1}{z^n} = 2\cos n\theta$

1

- (ii) For $z = r(\cos\theta + i\sin\theta)$, find r and the smallest positive value θ which satisfies $2z^3 = 9 + 3\sqrt{3}i$

2

- (b) (i) Find the values of A , B , and C such that:

3

$$\frac{4x^2 - 3x - 4}{x^3 + x^2 - 2x} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+2}$$

- (ii) Hence evaluate

$$\int \frac{4x^2 - 3x - 4}{x^3 + x^2 - 2x} dx$$

1

- (c) Solve the equation $x^4 - 7x^3 + 17x^2 - x - 26 = 0$, given that $x = (3 - 2i)$ is a root of the equation.

3

- (d) (i) Derive the equation of the tangent at the point $P\left(ct, \frac{c}{t}\right)$ on the rectangular hyperbola $xy = c^2$.

2

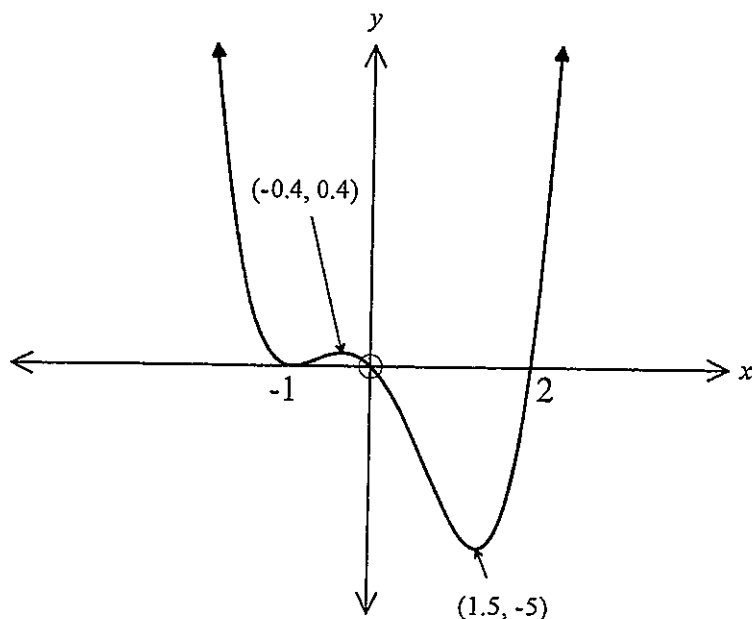
- (ii) Find the coordinates of A and B where this tangent cuts the x and y axis respectively.

2

- (iii) Prove that the area of the triangle OAB is a constant. (Where O is the origin).

1**End of Question 13**

- (a) The graph of $y = f(x)$ is shown below.

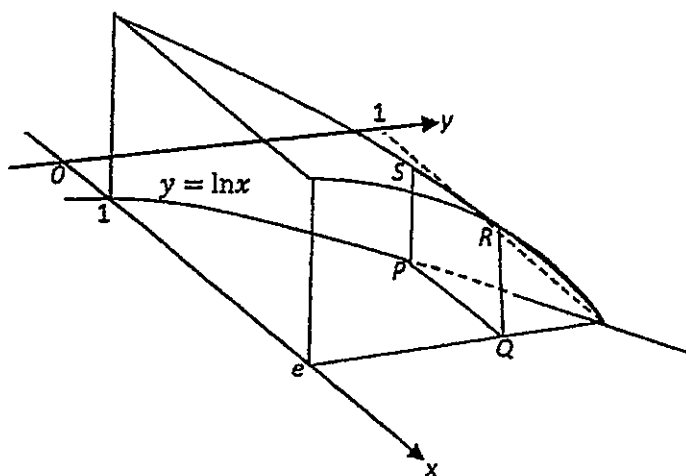


Draw neat, separate sketches for each of the following, showing all important features.

- | | | |
|-------|----------------------|---|
| (i) | $y = f(x) $ | 1 |
| (ii) | $y = \frac{1}{f(x)}$ | 2 |
| (iii) | $y^2 = f(x)$ | 2 |
| (iv) | $y = e^{f(x)}$ | 2 |
- (b) Show that the equation of the normal to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
- 3
- at the point $P(x_1, y_1)$ is given by the equation: $\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$
- (c) A particle is projected vertically upwards. The resistance to the motion is proportional to the square of the velocity. The velocity of projection is V
- | | | |
|------|---|---|
| (i) | Show that the acceleration is given by: $\ddot{x} = -(g + kv^2)$ | 1 |
| (ii) | Find the maximum height reached and the time taken to reach this height, expressing your answer in terms of V and k . | 4 |

End of Question 14

(a)



The base of a solid is the region bounded by the curve $y = \ln(x)$, the x -axis, and the lines $x = 1$ and $x = e$, as shown in the diagram.

Vertical cross-sections taken through this solid in a direction parallel to the x -axis are squares. A typical cross-section PQRS is shown.

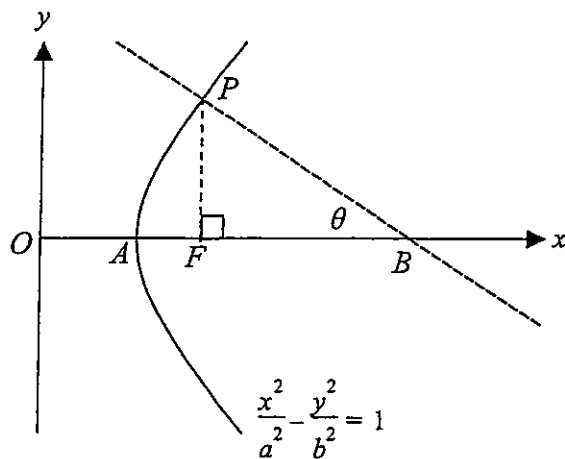
3

Find the volume of the solid.

- (b) A particle of mass m kg is dropped from rest in a medium where the resistance to motion has magnitude $\frac{1}{40}mv^2$ when the speed of the particle is v ms⁻¹. After t seconds the particle has fallen x metres. The acceleration due to gravity is 10 ms⁻².

- (i) Explain why $\ddot{x} = \frac{1}{40}(400 - v^2)$. 1
- (ii) Find an expression for t in terms of v by integration. 2
- (iii) Show that $v = 20\left(1 - \frac{2}{1 + e^t}\right)$. 1
- (iv) Find x as a function of t . 2

(c)



In the diagram, F is a focus of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ with eccentricity e .

This branch of the hyperbola cuts the x axis at A where $AF = h$. P is the point on the hyperbola vertically above F and the normal at P cuts the x axis at B making an acute angle θ with the x axis.

(i) Show that $\tan \theta = \frac{1}{e}$.

3

(ii) Show that $PF = h(e+1)$

1

(d) The equation $x^3 + 3x^2 + 2x + 1 = 0$ has roots α, β and γ .

Find the monic cubic equation with roots α^2, β^2 and γ^2 .

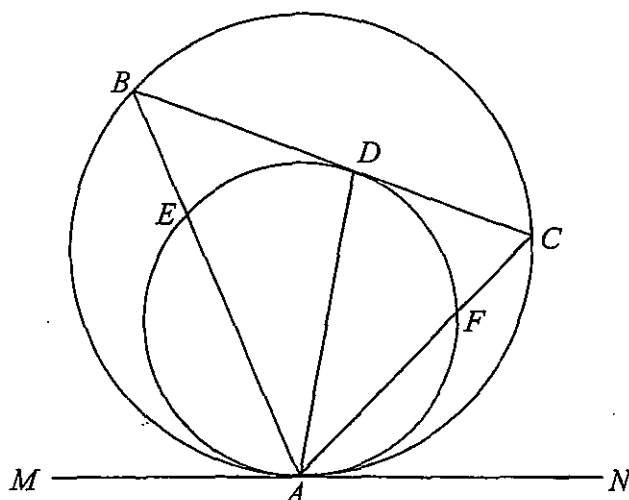
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End of Question 15

- (a) If $A(x)$ and $B(x)$ are odd polynomial functions show that the product $P(x) = A(x) \cdot B(x)$ is an even polynomial function.

2

(b)



In the diagram, MAN is the common tangent to two circles touching internally at A . B and C are two points on the larger circle such that BC is a tangent to the smaller circle with point of contact D . AB and AC cut the smaller circle at E and F respectively. Copy the diagram. Show that AD bisects $\angle BAC$.

4

- (c) Derive the reduction formula:

$$\int x^n e^{-x^2} dx = -\frac{1}{2} x^{n-1} e^{-x^2} + \frac{n-1}{2} \int x^{n-2} e^{-x^2} dx$$

and use this reduction formula to evaluate $\int_0^1 x^5 e^{-x^2} dx$

4

- (d) Use Mathematical Induction to prove that $5^n > 4n + 12$ for all integers $n > 1$.
- (e) Five letters are chosen from the word CHRISTMAS. These five letters are then placed alongside one another to form a five letter arrangement. Find the number of distinct arrangements that are possible, considering all choices.

3

2

End of Question 16

END OF EXAMINATION

page 12

Ext 2 (Solutions) TRIAL 2015

$$Q1/ \lim_{h \rightarrow 0} \frac{\sqrt{h+1} - 1}{h} \times \frac{\sqrt{h+1} + 1}{\sqrt{h+1} + 1}$$

$$= \lim_{h \rightarrow 0} \frac{h+1-1}{h(\sqrt{h+1}+1)}$$

$$= \lim_{h \rightarrow 0} \frac{1}{\sqrt{h+1}+1}$$

$$= \frac{1}{2} \quad (B)$$

$$Q2/ B-A = C-B$$

$$2B = A+C$$

$$B = \frac{A+C}{2}$$

$$\therefore 2B = A+C$$

$$= \frac{\sin 2B}{\sin B}$$

$$= \frac{2 \sin B \cos B}{\cancel{\sin B}}$$

$$= 2 \cos B \quad (A)$$

$$Q3/ x=1, x=-1 \text{ (vertical asymptotes)}$$

$$\lim_{x \rightarrow \infty} \frac{\frac{x^2}{x^2}}{\frac{x^2}{x^2} - \frac{1}{x^2}}$$

$$\lim_{x \rightarrow \infty} \frac{1}{1 - \frac{1}{x^2}}$$

$$= 1$$

$$\therefore y=1 \text{ (horizontal asymptote)}$$

$$(C)$$

$$4/ \quad z + \bar{z} = 2 \operatorname{Re} z \quad (D)$$

$$5/ \quad \text{Asymptotes are } y = \pm \frac{1}{\sqrt{3}} x$$

$$\therefore \tan \theta = \left| \frac{\frac{1}{\sqrt{3}} + \frac{1}{\sqrt{3}}}{1 - \frac{1}{3}} \right|$$

$$\tan \theta = \left| \frac{2}{\sqrt{3}} \div \frac{2}{3} \right|$$

$$\tan \theta = \frac{2}{\sqrt{3}} \times \frac{3}{2}$$

$$\tan \theta = \frac{3}{\sqrt{3}}$$

$$\therefore \tan \theta = \sqrt{3}$$

$$\theta = \frac{\pi}{3} \quad (C)$$

$$6/ \quad t = \tan \frac{x}{2}$$

$$\text{when } x = \frac{\pi}{2}$$

$$\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$$

$$t = \tan \frac{\pi}{4}$$

$$\therefore t = 1$$

$$\frac{2 dt}{\sec^2 \frac{x}{2}} = dx$$

$$\text{when } x = 0$$

$$t = \tan 0$$

$$\therefore t = 0$$

$$\frac{2 dt}{1+t^2} = dx$$

$$1 + \sin x = 1 + \frac{2t}{1+t^2}$$

$$\frac{1 + 2t}{1+t^2} = \frac{t^2 + 2t + 1}{1+t^2}$$

$$= \frac{(t+1)^2}{1+t^2}$$

$$\int_0^1 \frac{\cancel{t^2+1}}{(1+t)^2} \times \frac{2dt}{\cancel{t^2+1}}$$

$$= \int_0^1 \frac{2}{(1+t)^2} dt \quad (D)$$

$$7/ \quad y = 4x^2 - x^4$$

$$x^4 - 4x^2 + 4 = 4 - y$$

$$(x^2 - 2)^2 = 4 - y$$

$$(x^2 - 2) = \pm \sqrt{4 - y}$$

$$A = \pi (x_2^2 - x_1^2) \quad (\text{Radius outer} - \text{Radius inner})$$

$$\Delta V = \pi (x_2^2 - x_1^2) \Delta y$$

$$= \pi \left\{ (2 + \sqrt{4-y})^2 - (2 - \sqrt{4-y})^2 \right\} \Delta y$$

$$= 2\pi \sqrt{4-y} \Delta y$$

$$\therefore V = 2\pi \int_0^4 \sqrt{4-y} \, dy \quad (A)$$

$$8/ \quad x^4 + px + q = 0$$

$$\left. \begin{aligned} \alpha^4 + p\alpha + q &= 0 \\ \beta^4 + p\beta + q &= 0 \\ \gamma^4 + p\gamma + q &= 0 \\ \delta^4 + p\delta + q &= 0 \end{aligned} \right\} \text{ADD}$$

$$\alpha^4 + \beta^4 + \gamma^4 + \delta^4 + p(\alpha + \beta + \gamma + \delta) + 4q = 0$$

$$\alpha^4 + \beta^4 + \gamma^4 + \delta^4 + 0 + 4q = 0$$

$$\alpha^4 + \beta^4 + \gamma^4 + \delta^4 = -4q \quad (A)$$

$$9/ \quad \text{Domain is } -1 \leq x \leq 1$$

$$\therefore \sin^{-1}(-1) + \tan^{-1}(-1) \leq y \leq \sin^{-1}(1) + \tan^{-1}(1)$$

$$-\frac{\pi}{2} + -\frac{\pi}{4} \leq y \leq \frac{\pi}{2} + \frac{\pi}{4}$$

$$-\frac{3\pi}{4} \leq y \leq \frac{3\pi}{4} \quad (C)$$

$$10/ \quad e^x + \frac{dy}{dx} e^y = 0$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{-e^x}{e^y} \\ &= -e^{x-y} \end{aligned} \quad (A)$$

$$Q11 / (a) \quad z = 1 + 3i \quad \bar{z} = 1 - 3i$$

$$w = 2 - i \quad \bar{w} = 2 + i$$

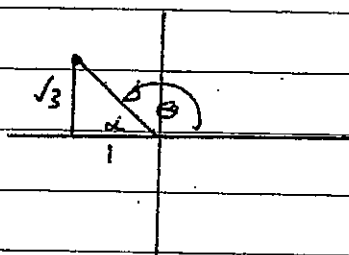
$$(i) \quad 1 - 3i - 2 + i = -1 - 2i$$

$$(ii) \quad (1 + 3i)(2 - i) = 2 - i + 6i + 3$$

$$= 5 + 5i$$

$$(b) (i) \quad R = \sqrt{3+1}$$

$$= 2$$



$$\tan \alpha = \sqrt{3}$$

$$\alpha = \frac{\pi}{3}$$

$$\therefore \theta = \frac{2\pi}{3}$$

$$\therefore -1 + \sqrt{3}i = 2 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)$$

$$= 2 \operatorname{cis} \frac{2\pi}{3}$$

$$(ii) \quad z^8 = 2^8 \left(\cos \frac{16\pi}{3} + i \sin \frac{16\pi}{3} \right)$$

$$= 2^8 \left(-\cos \frac{\pi}{3} - i \sin \frac{\pi}{3} \right)$$

$$= 2^8 \left(-\frac{1}{2} - \frac{\sqrt{3}i}{2} \right)$$

$$= 2^8 \underline{(-1 - \sqrt{3}i)}$$

$$= 2^7 \underline{2} (-1 - \sqrt{3}i)$$

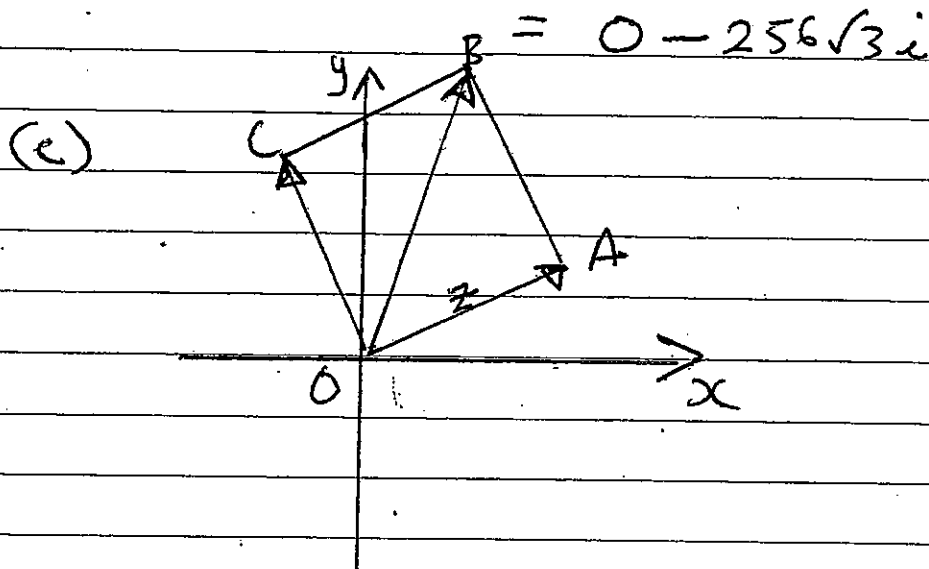
$$16z^4 = 16 \cdot 2^4 \left(\cos \frac{8\pi}{3} + i \sin \frac{8\pi}{3} \right)$$

$$= 2^8 \left(-\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$$

$$= 2^8 \left(-\frac{1}{2} + \frac{\sqrt{3}i}{2} \right)$$

$$= 2^7 (-1 + \sqrt{3}i)$$

$$\begin{aligned}
 \therefore z^8 - 16z^4 &= 2^7(-1-\sqrt{3}i) - 2^7(-1+\sqrt{3}i) \\
 &= -2^7 - 2^7\sqrt{3}i + 2^7 - 2^7\sqrt{3}i \\
 &= -2 \cdot 2^7\sqrt{3}i \\
 &= -2^8\sqrt{3}i
 \end{aligned}$$



$$\overrightarrow{OA} = z \quad \therefore \overrightarrow{OC} = iz$$

$$\begin{aligned}
 \text{(i)} \quad \overrightarrow{OB} &= \overrightarrow{OA} + \overrightarrow{OC} \\
 &= z + iz \\
 &= z(1+i)
 \end{aligned}$$

$$\text{(ii)} \quad 4 + 2i = z(1+i)$$

$$\frac{4+2i}{1+i} = z$$

$$\frac{(4+2i)}{(1+i)} \times \frac{(1-i)}{(1-i)} = z$$

$$6 - 2i = 2z$$

$$3 - i = z$$

$$(d) \quad P(x) = x^6 + ax^3 + bx^2$$

$$P(-1) = 0$$

$$\therefore 0 = (-1)^6 + a(-1)^3 + b(-1)^2$$

$$0 = 1 - a + b$$

$$\boxed{a - b = 1}$$

$$P'(x) = 6x^5 + 3ax^2 + 2bx$$

$$P'(-1) = 0$$

$$\therefore 0 = 6(-1)^5 + 3a(-1)^2 + 2b(-1)$$

$$0 = -6 + 3a - 2b$$

$$\boxed{3a - 2b = 6}$$

$$\left. \begin{array}{l} 3a - 2b = 6 \\ 2a - 2b = 2 \end{array} \right\} \text{ subtract}$$

$$a = 4$$

$$b = 3$$

(e) sum of roots 1 at a time are

$$\alpha + \frac{1}{\alpha} + \beta + \frac{1}{\beta} = -\frac{b}{a}$$

$$\therefore \alpha + \frac{1}{\alpha} + \beta + \frac{1}{\beta} = -b$$

Sum of roots 3 at a time

$$(\alpha \times \frac{1}{\alpha} \times \beta) + (\alpha \times \beta \times \frac{1}{\beta}) + (\frac{1}{\alpha} \times \beta \times \frac{1}{\beta})$$

$$+ (\alpha \times \frac{1}{\alpha} \times \frac{1}{\beta}) = -\frac{d}{a}$$

$$\beta + \alpha + \frac{1}{\alpha} + \frac{1}{\beta} = -d$$

$$\therefore b = d$$

$$Q12/(a) \int \frac{e^{-x}}{1+e^x} dx$$

$$= \int \frac{1}{e^x(1+e^x)} dx$$

For partial fractions let $u = e^x$

$$\frac{1}{u(1+u)} = \frac{a}{u} + \frac{b}{(1+u)}$$

$$\therefore 1 \equiv a(1+u) + bu$$

$$\text{Let } u = 0$$

$$a = 1$$

$$\text{Let } u = -1$$

$$b = -1$$

$$\therefore \int \frac{1}{e^x(1+e^x)} dx = \int \frac{1}{e^x} - \frac{1}{1+e^x} dx$$

$$= \int e^{-x} - \left(\frac{1+e^x - e^x}{1+e^x} \right) dx$$

$$= \int e^{-x} - \left(1 - \frac{e^x}{1+e^x} \right) dx$$

$$= -e^{-x} - x + \ln(1+e^x) + C$$

$$(ii) \int \frac{x^2}{x+1} dx$$

$$\begin{array}{r}
 x-1 \\
 x+1 \overline{) x^2+0x+0} \\
 \underline{x^2+x} \\
 -x+0 \\
 \underline{-x-1} \\
 1
 \end{array}$$

$$\int \frac{x^2}{x+1} dx = \int x-1 + \frac{1}{x+1} dx$$

$$= \frac{x^2}{2} - x + \ln|x+1| + C$$

$$(b) \int_0^{\frac{\pi}{2}} \sec^3 2\theta d\theta$$

$$= \int_0^{\frac{\pi}{2}} \sec^2 2\theta \cdot \sec 2\theta d\theta$$

$$\text{Let } u = \sec 2\theta$$

$$u = (\cos 2\theta)^{-1}$$

$$\frac{du}{d\theta} = -(\cos 2\theta)^{-2} \cdot -2\sin 2\theta$$

$$= \frac{2\sin 2\theta}{\cos 2\theta \cdot \cos 2\theta}$$

$$= 2 \sec 2\theta \tan 2\theta$$

$$\frac{dv}{d\theta} = \sec^2 \theta$$

$$v = \frac{\tan 2\theta}{2}$$

$$= \left[\frac{\tan 2\theta}{2} \cdot \sec 2\theta \right]_0^{\frac{\pi}{6}} - \int_0^{\frac{\pi}{6}} \sec 2\theta \cdot \tan^2 2\theta d\theta$$

$$= \left[\frac{\tan \frac{\pi}{3}}{2} \cdot \sec \frac{\pi}{3} \right] - \int_0^{\frac{\pi}{6}} \sec 2\theta (\sec^2 2\theta - 1) d\theta$$

$$= \frac{\sqrt{3}}{2} \cdot 2 - \int_0^{\frac{\pi}{6}} \sec^3 2\theta - \sec 2\theta d\theta$$

$$\therefore 2 \int_0^{\frac{\pi}{6}} \sec^3 2\theta d\theta = \sqrt{3} + \int_0^{\frac{\pi}{6}} \sec 2\theta d\theta$$

$$= \sqrt{3} + \int_0^{\frac{\pi}{6}} \frac{\sec 2\theta (\sec 2\theta + \tan 2\theta)}{(\sec 2\theta + \tan 2\theta)} d\theta$$

$$= \sqrt{3} + \int_0^{\frac{\pi}{6}} \frac{\sec^2 2\theta + \sec 2\theta \tan 2\theta}{(\sec 2\theta + \tan 2\theta)} d\theta$$

$$= \sqrt{3} + \frac{1}{2} \int_0^{\frac{\pi}{6}} \frac{2\sec^2 2\theta + 2\sec 2\theta \tan 2\theta}{(\sec 2\theta + \tan 2\theta)} d\theta$$

$$= \sqrt{3} + \frac{1}{2} \ln \left[\sec 2\theta + \tan 2\theta \right] \Big|_0^{\frac{\pi}{6}}$$

$$= \sqrt{3} + \frac{1}{2} \ln \left[\sec \frac{\pi}{3} + \tan \frac{\pi}{3} \right] - 0$$

$$= \sqrt{3} + \frac{1}{2} \ln (2 + \sqrt{3})$$

$$\int_0^{\frac{\pi}{6}} \sec^3 2\theta \, d\theta = \frac{\sqrt{3}}{2} + \frac{1}{4} \ln (2 + \sqrt{3})$$

(c) $x = \tan \theta$
 $\frac{dx}{d\theta} = \sec^2 \theta$
 $dx = \sec^2 \theta \, d\theta$

when $x = \sqrt{3}$
 $\sqrt{3} = \tan \theta$
 $\theta = \frac{\pi}{3}$

when $x = 1$
 $1 = \tan \theta$
 $\theta = \frac{\pi}{4}$

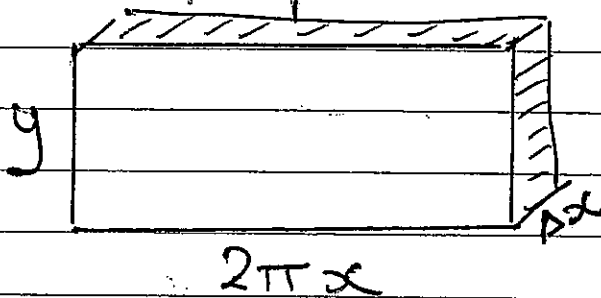
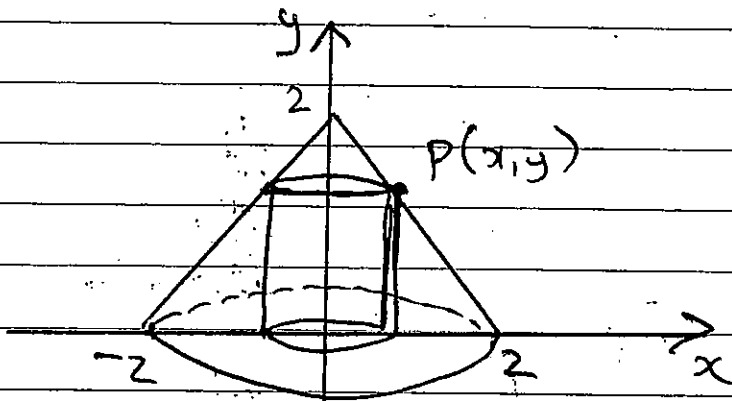
$$\int_1^{\sqrt{3}} \frac{1}{x^2 \sqrt{1+x^2}} \, dx = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{1}{\tan^2 \theta \cdot \sec \theta} \cdot \sec^2 \theta \, d\theta$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sec \theta}{\tan^2 \theta} \, d\theta$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{1}{\cancel{\cos \theta} \times \frac{\sin \theta}{\cancel{\cos \theta}} \times \tan \theta} d\theta$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \operatorname{cosec} \theta \cot \theta d\theta$$

(d)



$$\Delta V = 2\pi xy \Delta x$$

$$\text{But } y = (2-x)$$

$$\Delta V = 2\pi (2x - x^2) \Delta x$$

$$\text{Total Volume} = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^2 2\pi (2x - x^2) \Delta x$$

$$= 2\pi \int_0^2 (2x - x^2) dx$$

$$= 2\pi \left[x^2 - \frac{x^3}{3} \right]_0^2$$

$$= 2\pi \left[4 - \frac{8}{3} - (0 - 0) \right]$$

$$= 2\pi \left[\frac{4}{3} \right]$$

$$= \frac{8\pi}{3} u^2$$

(13)

$$z = \cos \theta + i \sin \theta$$

$$z^n = \cos(n\theta) + i \sin(n\theta)$$

$$\frac{1}{z^n} = z^{-n} = \cos(-n\theta) + i \sin(-n\theta)$$

$$= \cos(n\theta) - i \sin(n\theta)$$

$$z^n + \frac{1}{z^n} = \cos(n\theta) + i \sin(n\theta) + \cos(n\theta) - i \sin(n\theta)$$

$$= 2 \cos(n\theta)$$

$$(ii) z^3 = r^3 (\cos 3\theta + i \sin 3\theta)$$

$$2r^3 (\cos 3\theta + i \sin 3\theta) = 9 + 3\sqrt{3}i$$

$$\therefore 2r^3 \cos 3\theta = 9 \quad \text{and} \quad 2r^3 \sin 3\theta = 3\sqrt{3}$$

by division

$$\tan 3\theta = \frac{3\sqrt{3}}{9}$$

$$= \frac{\sqrt{3}}{3} \quad \text{OR} \quad \frac{1}{\sqrt{3}}$$

$$\therefore 3\theta = \frac{\pi}{6} \quad (\text{smallest positive value})$$

$$\theta = \frac{\pi}{18}$$

Sub into $2r^3 \cos 3\theta = 9$

$$2r^3 \cos \frac{\pi}{6} = 9$$

$$\frac{2r^3 \sqrt{3}}{2} = 9$$

$$r^3 = \frac{9}{\sqrt{3}}$$

$$r^3 = \frac{9}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$r^3 = 3\sqrt{3}$$

$$r = \sqrt{3}$$

$$(b)(i) A(x-1)(x+2) + Bx(x+2) + C(x-1)x \equiv 4x^2 - 3x - 4$$

$$\text{Let } x=0 \quad -2A = -4 \\ A = 2$$

$$\text{Let } x=1 \quad 3B = -3 \\ B = -1$$

$$\text{Let } x=-2 \quad 6C = 18 \\ C = 3$$

$$(ii) \int \frac{2}{x} + \frac{-1}{x-1} + \frac{3}{x+2} dx$$

$$= 2 \ln x - \ln(x-1) + 3 \ln(x+2) + C$$

(c) As $(3-2i)$ is a factor then $3+2i$ is also a factor since the coefficients are real.

$$\therefore x^2 - (3-2i + 3+2i)x + (3-2i)(3+2i) \\ = x^2 - 6x + 13 \quad \text{is a factor}$$

$$\begin{array}{r} x^2 - x - 2 \\ x^2 - 6x + 13 \end{array} \overline{) x^4 - 7x^3 + 17x^2 - x - 26}$$

$$\underline{x^4 - 6x^3 + 13x^2}$$

$$-x^3 + 4x^2 - x$$

$$\underline{-x^3 + 6x^2 - 13x}$$

$$-2x^2 + 12x - 26$$

$$\underline{-2x^2 + 12x - 26}$$

0

$\therefore x^2 - x - 2$ is a factor
 $\therefore (x+1)(x-2)$ are factors

$\therefore$ Solutions to $x^4 - 7x^3 + 17x^2 - x - 26 = 0$
 are $x = 3 \pm 2i, -1$ and 2

(d) $xy = c^2 \quad y = c^2 x^{-1}$

$$\frac{dy}{dx} = \frac{-c^2}{x^2} \quad \text{at } x = ct$$

$$m_T = \frac{-c^2}{c^2 + 2}$$

$$= -\frac{1}{1+2}$$

$$y - \frac{c}{1+2} = -\frac{1}{1+2} (x - ct)$$

$$1+2 y - ct = -x + ct$$

$$x + 1+2 y - 2ct = 0$$

(iii) When $x = 0 \quad 1+2 y = 2ct$

$$y = \frac{2ct}{1+2}$$

$$y = \frac{2c}{1+2}$$

$\therefore B$ is $(0, \frac{2c}{1+2})$

When $y=0$ $x=2ct$

$\therefore$ A is $(2ct, 0)$

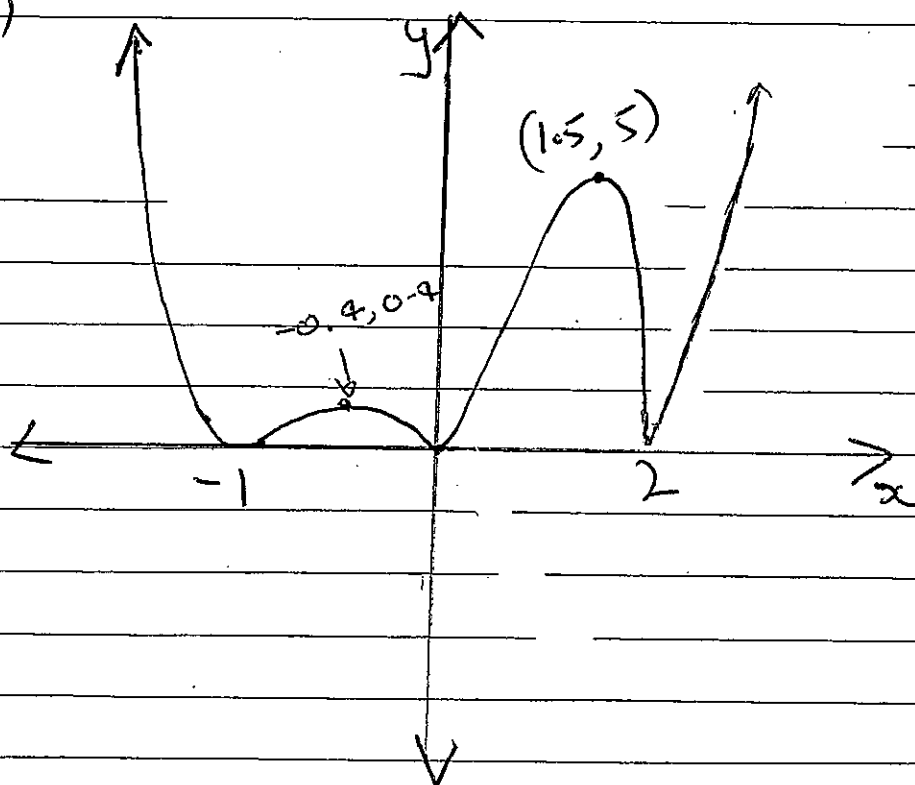
(iii) $OA = 2ct$
 $OB = \frac{2c}{t}$

$$\text{Area } \triangle OAB = \frac{1}{2} \times 2ct \times \frac{2c}{t}$$

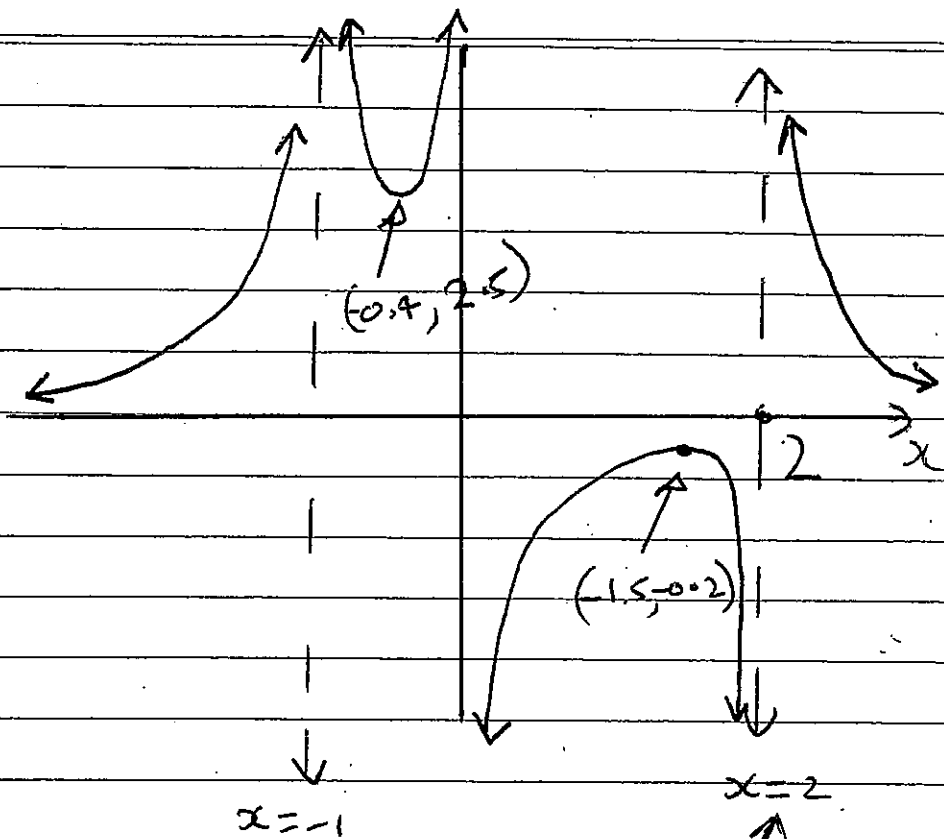
$$= 2c^2$$

which is constant as c is
a constant.

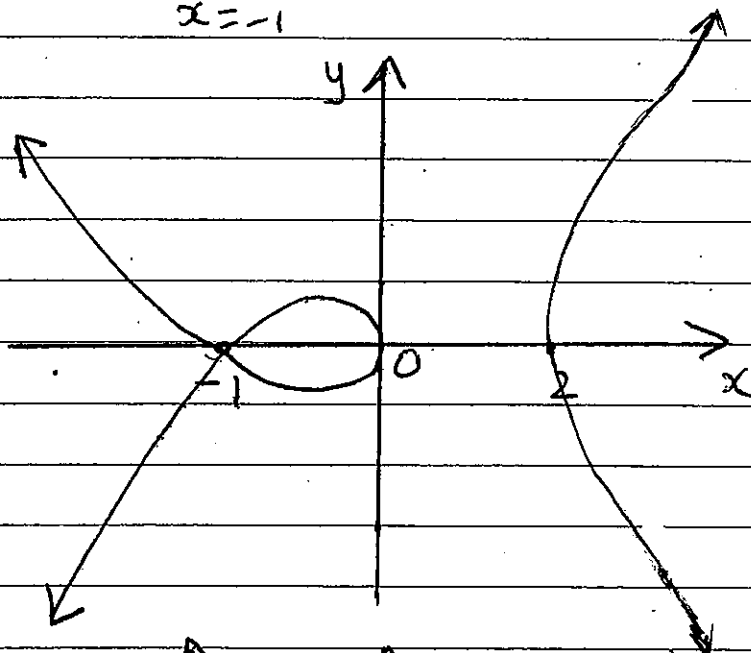
Q19(a)



(ii)

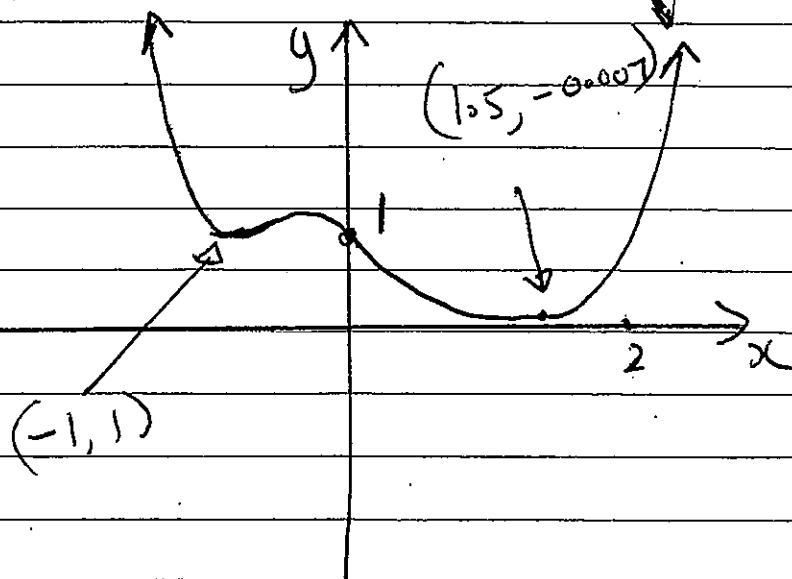


iii



concave up
at ∞

(iv)



$$(b) \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} + \frac{dy}{dx} \frac{2y}{b^2} = 0$$

$$\frac{dy}{dx} = -\frac{2x}{a^2} \times \frac{b^2}{2y}$$

$$= -\frac{b^2 x}{a^2 y}$$

$$\text{At } (x_1, y_1) \quad m_n = \frac{a^2 y_1}{b^2 x_1}$$

$$\therefore y - y_1 = \frac{a^2 y_1}{b^2 x_1} (x - x_1)$$

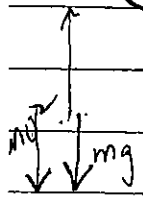
$$b^2 x, y - b^2 x_1, y_1 = a^2 y_1 x - a^2 x_1 y_1$$

Divide by $x_1 y_1$

$$\frac{b^2 y}{y_1} - b^2 = \frac{a^2 x}{x_1} - a^2$$

$$\therefore \frac{a^2 x}{x_1} - \frac{b^2 y}{y_1} = a^2 - b^2$$

(on rearranging)



$$(c) \quad m \ddot{x} = -mg - mkv^2$$

$$\ddot{x} = -g - kv^2$$

Taking upwards
as positive

$$\therefore \ddot{x} = -(g + kv^2)$$

$$(ii) \quad v \frac{dv}{dx} = -(g + kv^2)$$

$$\frac{dv}{dx} = \frac{-(g + kv^2)}{v}$$

$$\frac{dx}{dv} = -\frac{v}{g + kv^2}$$

$$x = - \int \frac{v}{g + kv^2} dv$$

$$x = -\frac{1}{2k} \int \frac{2kv}{g + kv^2} dv$$

$$x = -\frac{1}{2k} \ln(g + kv^2) + C$$

When $x=0$ $v=V$

$$0 = -\frac{1}{2k} \ln(g + kV^2) + C$$

$$\therefore C = \frac{1}{2k} \ln(g + kV^2)$$

$$x = -\frac{1}{2k} \ln(g + kv^2) + \frac{1}{2k} \ln(g + kV^2)$$

$$x = \frac{1}{2k} \ln \left\{ \frac{g + kV^2}{g + kv^2} \right\}$$

Maximum height when $v = 0$

$$x_{\max} = \frac{1}{2k} \ln \left(\frac{g + kV^2}{g} \right)$$

$$\frac{dv}{dt} = -(g + kv^2)$$

$$\frac{dt}{dv} = \frac{-1}{g + kv^2}$$

$$t = - \int \frac{1}{g + kv^2} dv$$

$$t = -\frac{1}{k} \int \frac{1}{\frac{g}{k} + v^2} dv$$

$$\therefore t = -\frac{1}{k} \times \frac{1}{\sqrt{\frac{g}{k}}} \tan^{-1} \left(\frac{\sqrt{k} v}{\sqrt{g}} \right) + C$$

$$t = -\frac{1}{k} \frac{\sqrt{k}}{\sqrt{g}} \tan^{-1} \left(\frac{\sqrt{k} v}{\sqrt{g}} \right) + C$$

when $t=0$ $v=V$

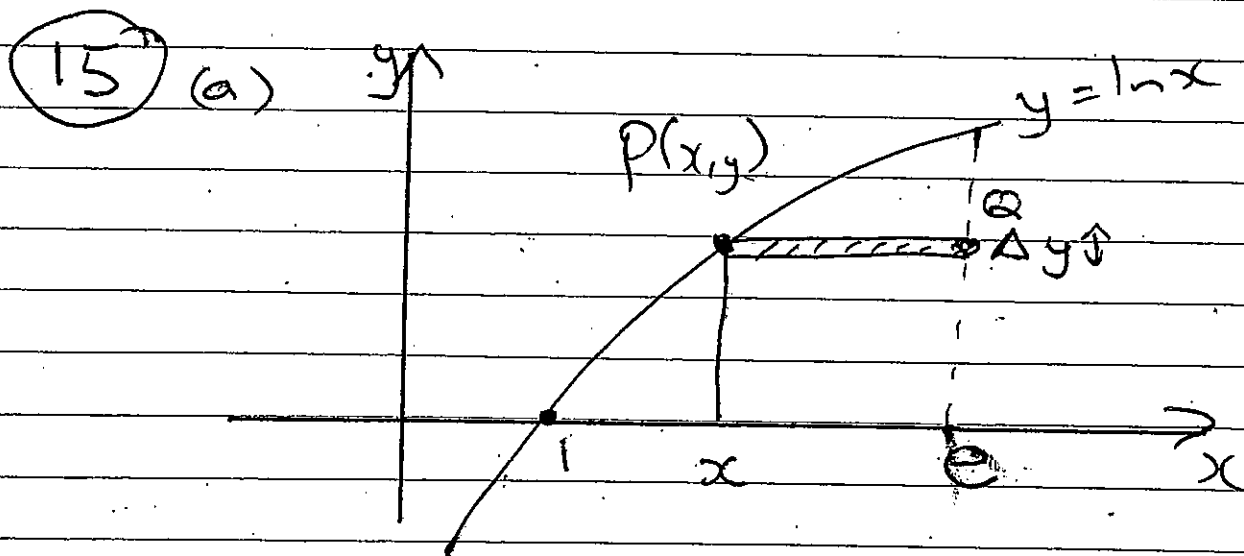
$$0 = -\frac{1}{\sqrt{Kg}} \tan^{-1} \left(\frac{\sqrt{K}}{\sqrt{g}} v \right) + C$$

$$C = \frac{1}{\sqrt{Kg}} \tan^{-1} \left(\frac{\sqrt{K}}{\sqrt{g}} V \right)$$

$$\therefore t = -\frac{1}{\sqrt{Kg}} \tan^{-1} \left(\frac{\sqrt{K}}{\sqrt{g}} v \right) + \frac{1}{\sqrt{Kg}} \tan^{-1} \left(\frac{\sqrt{K}}{\sqrt{g}} V \right)$$

Max height when $v=0$

$$\text{i.e. } t = \frac{1}{\sqrt{Kg}} \tan^{-1} \left(\frac{\sqrt{K}}{\sqrt{g}} V \right)$$



Take a slice through $P(x, y)$ parallel to the x axis with thickness Δy

$$PQ = e - x$$

$$\therefore \text{Area of square PQRS} = (e - x)^2$$

$$\therefore \Delta V = (e - x)^2 \Delta y$$

Total

$$\text{Volume} = \lim_{\Delta y \rightarrow 0} \sum_{x=1}^e (e - x)^2 \Delta y$$

$$\text{but } y = \ln x$$

$$\frac{dy}{dx} = \frac{1}{x}$$

$$dy = \frac{1}{x} dx$$

$$\therefore V = \int_1^e (e - x)^2 \cdot \frac{1}{x} dx$$

$$V = \int_1^e (e^2 - 2ex + x^2) \times \frac{1}{x} dx$$

$$= \int_1^e \left(\frac{e^2}{x} - 2e + x \right) dx$$

$$= \left[e^2 \ln x - 2ex + \frac{x^2}{2} \right]_1^e$$

$$= e^2 - 2e^2 + \frac{e^2}{2} - \left(0 - 2e + \frac{1}{2} \right)$$

$$= -\frac{e^2}{2} + 2e - \frac{1}{2}$$

Taking Downwards as +

Q15/(b)

(i)

$$\begin{aligned} t &= 0 \\ x &= 0 \\ v &= 0 \end{aligned}$$

$\downarrow x$

Forces on particle

$\uparrow \frac{1}{40} v^2$
 $\downarrow mg$

$$m \ddot{x} = mg - \frac{1}{40} m v^2$$

$$\ddot{x} = \frac{1}{40} (400 - v^2) \quad \text{as } g = 10$$

$$(ii) \quad \frac{dv}{dt} = \frac{1}{40} (400 - v^2)$$

$$\frac{dt}{dv} = \frac{40}{20^2 - v^2}$$

$$= \frac{1}{20+v} + \frac{1}{20-v}$$

$$t = \ln \left(\frac{20+v}{20-v} \right) + c$$

$$\text{When } v=0, t=0 \therefore c=0$$

$$t = \ln \left(\frac{20+v}{20-v} \right)$$

$$(iii) \quad \text{as } t = \ln \left(\frac{20+v}{20-v} \right)$$

$$\text{Then } e^t = \frac{20+v}{20-v}$$

$$(20-v)e^+ = 20+v$$

$$20e^+ - ve^+ = 20+v$$

$$20e^+ - 20 = v + ve^+$$

$$20(e^+ - 1) = v(1 + e^+)$$

$$\frac{20(e^+ - 1)}{1 + e^+} = v$$

$$20 \left(\frac{1 + e^+}{1 + e^+} - \frac{2}{1 + e^+} \right) = v$$

$$\therefore v = 20 \left(1 - \frac{2}{1 + e^+} \right)$$

$$(iv) \frac{dx}{dt} = 20 \left(1 - \frac{\frac{2}{e^+}}{\frac{1}{e^+} + \frac{e^+}{e^+}} \right)$$

$$= 20 \left(1 - \frac{2e^{-+}}{e^{-+} + 1} \right)$$

$$x = 20 \left(t - 2 \ln(1 + e^{-+}) \right)$$

$$\text{When } x=0, t=0$$

$$0 = 20(0 - 2 \ln 2)$$

$$= -40 \ln 2$$

$$\therefore x = 20 \left(t - 2 \ln \left(\frac{1 + e^{-t}}{2} \right) \right)$$

(c) F is $(ae, 0)$ A is $(a, 0)$

$$b^2 = a^2(e^2 - 1)$$

$$PF = e \cdot PM$$

$$= e \left(ae - \frac{a}{e} \right)$$

$$= ae^2 - a$$

$$= a(e^2 - 1) \quad \therefore P \text{ is } (ae, a(e^2 - 1))$$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} - \frac{dy}{dx} \frac{2y}{b^2} = 0$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{2x}{a^2} \times \frac{b^2}{2y} \\ &= \frac{b^2 x}{a^2 y} \end{aligned}$$

$$\therefore m_n = -\frac{a^2 y}{b^2 x} \quad \text{so at } P$$

$$m_n = -\frac{a^2 \cdot a(e^2 - 1)}{b^2 ae}$$

$$\text{but } b^2 = a^2(e^2 - 1)$$

$$m_n = -\frac{1}{e}$$

$$\text{then } \tan(180 - \theta) = m_n$$

$$\therefore \tan \theta = \frac{1}{e}$$

$$(ii) \quad h = AF = a(e-1)$$

$$\text{and } PF = a(e^2-1)$$

$$\therefore h(e+1) = a(e-1)(e+1) \\ = a(e^2-1)$$

$$\therefore PF = h(e+1)$$

$$(d) \quad x^3 + 3x^2 + 2x + 1 = 0$$

$\therefore$ with roots α^2 , β^2 and γ^2 we get

$$(x^{\frac{1}{2}})^3 + 3(x^{\frac{1}{2}})^2 + 2(x^{\frac{1}{2}}) + 1 = 0$$

$$x^{\frac{3}{2}} + 3x + 2x^{\frac{1}{2}} + 1 = 0$$

$$x^{\frac{3}{2}} + 2x^{\frac{1}{2}} = -(3x+1)$$

$$x^{\frac{1}{2}}(x+2) = -(3x+1)$$

Square both sides,

$$x(x+2)^2 = (3x+1)^2$$

$$x(x^2 + 4x + 4) = 9x^2 + 6x + 1$$

$$x^3 + 4x^2 + 4x = 9x^2 + 6x + 1$$

$\therefore$ monic equation is

$$x^3 - 5x^2 - 2x - 1 = 0$$

Q16/ (a) As both $A(x)$ and $B(x)$ are odd

then $A(-x) = -A(x)$

$$B(-x) = -B(x)$$

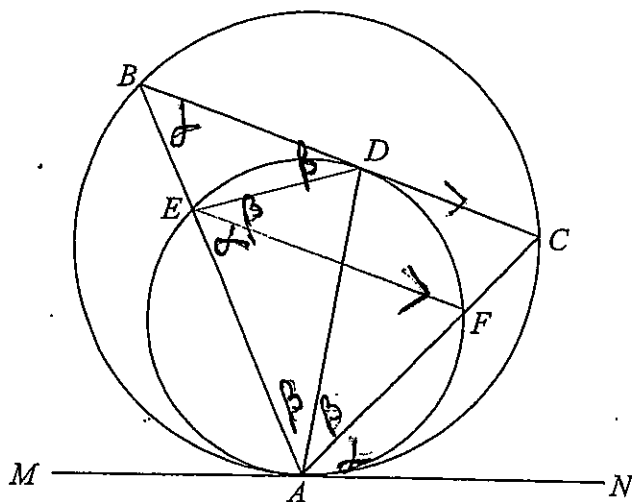
Now $P(-x) = A(-x) \cdot B(-x)$

for $P(x) = A(x) \cdot B(x)$

$$\therefore P(-x) = -A(x) - B(x)$$
$$= A(x) \cdot B(x)$$

$\therefore P(-x) = P(x) \therefore \text{even}$

(b)



Construct ED and EF

$\angle CAN = \angle ABC$ (alternate large circle segment theorem)

$$\angle FAN = \angle AEF \text{ (alternate segment theorem)}$$
$$\angle ABC = \angle EFG$$

$\therefore BC \parallel EF$ (Corresponding $\angle$'s equal)

$\angle BDE = \angle DAE = \beta$ (alternate segment theorem large circle)

$\angle BDE = \angle DEF = \beta$ (alternate $\angle$'s $BC \parallel EF$)

$\angle DEF = \angle DAC = \beta$ (equal $\angle$'s on circumference standing on arc DF)

$$\therefore \angle DAE = \angle DAC = \beta$$

Hence AD bisects $\angle BAC$

(c) $\int x^n e^{-x^2} dx$

$$\text{let } u = x^{n-1}$$

$$\frac{du}{dx} = (n-1)x^{n-2}$$

$$v' = x e^{-x^2}$$

$$v = -\frac{1}{2} e^{-x^2}$$

$$\therefore \int x^n e^{-x^2} dx = -\frac{1}{2} x^{n-1} e^{-x^2} + \frac{n-1}{2} \int x^{n-2} e^{-x^2} dx$$

$$\int_0^1 x^5 e^{-x^2} dx = \left[-\frac{1}{2} x^4 e^{-x^2} \right]_0^1 + 2 \int_0^1 x^3 e^{-x^2} dx$$

$$= -\frac{1}{2e} + 2 \left[-\frac{1}{2} x^2 e^{-x^2} \right]_0^1 + 1 \int_0^1 x e^{-x^2} dx$$

$$= -\frac{1}{2e} + 2x - \frac{1}{2e} + 2 \left[-\frac{1}{2} x^0 e^{-x^2} \right]_0^1$$

$$= -\frac{1}{2e} - \frac{1}{e} + 2 \left[-\frac{1}{2e} - \frac{1}{2} \right]$$

$$= -\frac{1}{2e} - \frac{1}{e} - \frac{1}{e} - 1$$

$$= -\frac{1}{2e} - \frac{2}{e} - 1$$

$$= -\frac{1}{2e} - \frac{4}{2e} - 1$$

$$= -1 - \frac{5}{2e}$$

(d) $5^n > 4n + 12$ for $n > 1$

Step 1 Prove true for $n=2$

$$5^2 > 4(2) + 12$$

$$25 > 20 \quad \therefore \text{true for } n=2$$

Step 2 Assume true for $n=k$

where k is a positive integer

i.e. $5^k > 4k + 12$

Step 3 Prove true for $n=k+1$

i.e. $5^{k+1} > 4k + 16$

$$\text{LHS} = 5 \cdot 5^k$$

$$> 5(4k + 12)$$

from assumption

$$> 20k + 60 > 4k + 16 \quad \text{which is true for } k \text{ being a positive integer.}$$

Step 4. As it is true for $n=2$ and if true for $n=k$, it is true for $n=k+1$, therefore true for all positive integers of n , $n > 1$

(e) 2 S's

$$\boxed{7} \boxed{6} \boxed{5} \times \frac{20}{2!} \text{ ways} = 2100 \text{ arrangements}$$

1 S

$$\boxed{7} \boxed{6} \boxed{5} \boxed{4} \times 5 \text{ ways} = 4200 \text{ arrangements}$$

No S's

$$\boxed{7} \boxed{6} \boxed{5} \boxed{4} \boxed{3} = 2520 \text{ arrangements}$$

∴ Total

$$\text{arrangements} = 2100 + 4200 + 2520$$

$$= 8820 \text{ arrangements}$$

Alternative approach

3 Types	No of Selections	x	No of arrangements	
2 S's	$(1 \times {}^7C_3)$	$\times \frac{5!}{2!}$	$= 2100$	
1 S	$(1 \times {}^7C_4)$	$\times 5!$	$= 4200$	
0 S's	7C_5	$\times 5!$	$= 2520$	
			<u>8820</u>	